

AS3: Scrape hierarchical features

Part 2

- Scrape **several features** from a single page
- Challenging level: ★ ★
- A naïve idea: Independent selectors in parallel, each for one feature in the parse function

```
def parse(self, response):  
    quotes = response.xpath('//div[@class="post quote"]/span/text()).extract()
```

```
def parse(self, response):  
    authors = response.xpath('//div[@class="post quote" ]/div[@class="source"]/text()).extract()
```

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def parse(self, response):  
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```

How independent selectors work

ONE QUOTE PER DAY.

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So long...

...and thanks for all the fish.

Today's quote was the last one – after 12 years (and some change), I am closing this blog down. As in, there won't be any new posts, but you can still hit the ["Random"](#) button each day to read one (or more) of the 4395 quotes I've collected. There's definitely a bunch of duplicates but

1, 2023 / 15 notes

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```

➤ `<div class="post quote">`

➤ `<span>QUOTE</span>`

➤ `<div class="source">AUTHOR</div>`

➤ `<div class="tags">`

➤ ...

Quote 1	Author 1
Quote 2	Author 2
Quote 3	Author 3
.....	

"A book of quotations ... can never be complete."

—Robert M. Hamilton

#Hamilton #quote #quotes

January 1, 2023 / 10 notes

"Cheers to a new year and another chance for us to get it right."

—Oprah Winfrey

#Winfrey #quote #quotes

December 31, 2022 / 7 notes

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When independent selectors create trouble

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➤ `<div class="post quote">`

➤ `<span>QUOTE</span>`

➤ `<div class="source">AUTHOR</div>`

➤ `<div class="tags">`

➤ ...

➤ `<div class="post quote">`

➤ `<span>QUOTE</span>`

➤ `<div class="tags">`

➤ ...

Quote 1	Author 1	"A book of quotations ... can never be complete."
Quote 2	Author 3	—Robert M. Hamilton
Quote 3	Author 4	#Hamilton #quote #quotes
.....		January 1, 2023 / 10 notes
		"Cheers to a new year and another chance for us to get it right."
		#Winfrey #quote #quotes
		December 31, 2022 / 7 notes

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    authors = response.xpath('///div[@class="post quote" ]/div[@class="source"]/text()').extract()
```

➤ `<div class="post quote">`

➤ `<span>QUOTE</span>`

➤ `<div class="source">AUTHOR</div>`

➤ `<div class="tags">`

➤ ...

➤ `<div class="post quote">`

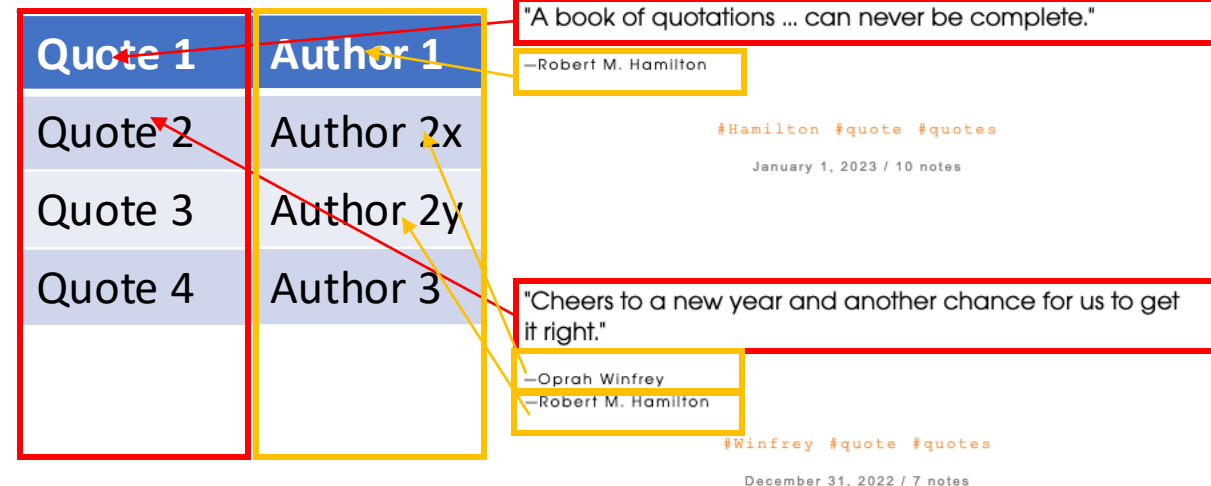
➤ `<span>QUOTE</span>`

➤ `<div class="source">AUTHOR 1</div>`

➤ `<div class="source">AUTHOR 2</div>`

➤ `<div class="tags">`

➤ ...



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Problem with independent selectors

Quote 1	Quote 2	Quote 3	Quote 4		Quote n-1	Quote n
Author 1	Author 2	Author 3	Author 4		Author n-1	Author n

- `<div class="post quote">`
 - `<span>QUOTE</span>`
 - `<div class="source">AUTHOR</div>`
 - `<div class="tags">`
 - ...

Author 1	Author 3	Author 4	Author 5		Author n
----------	----------	----------	----------	-------	----------

- `<div class="post quote">`
 - `<span>QUOTE</span>`
 - `<div class="tags">`
 - ...

Author 1	Author 2x	Author 2y	Author 3		Author n-2	Author n-1	Author n
----------	-----------	-----------	----------	-------	------------	------------	----------

- `<div class="post quote">`
 - `<span>QUOTE</span>`
 - `<div class="source">AUTHOR1</div>`
 - `<div class="source">AUTHOR2</div>`
 - `<div class="tags">`
 - ...

How to avoid misalignment

- Fill in one row after another
- In each row = one wrapper
 - Fill in column 1 (a quote)
 - Fill in column 2 (an author)
 -

```
▼ <div id="entries">
  ▶ <div class="post text">⋮</div>
  ➔ <div class="post quote">
    ➔ <span>"A book of quotations ... can never be complete."</span>
    ➔ <div class="source">Robert M. Hamilton</div>
    ▶ <div class="tags">⋮</div>
    ▶ <div class="post_footer">⋮</div>
    <!-- Post Footer -->
  </div>
  ➔ <div class="post quote">
    ➔ <span>"Cheers to a new year and another chance for us to get it right."</span>
    ➔ <div class="source">Oprah Winfrey</div>
    ▶ <div class="tags">⋮</div>
    ▶ <div class="post_footer">⋮</div>
    <!-- Post Footer -->
  </div>
  <div class="post quote">⋮</div>
  <div class="post quote">⋮</div>
  <div class="post quote">⋮</div>
  <div class="post quote">⋮</div>
```

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January 1, 2023 / 15 notes

"A book of quotations ... can never be complete."

—Robert M. Hamilton

#Hamilton #quote #quotes

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"Cheers to a new year and another chance for us to get it right."

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Build a hierarchical order

#The path to a particular wrapper

#Only extract the first value

- Find all the wrappers
- Write a 'for' loop to work on one wrapper a time
 - In one wrapper, get any values requested (e.g., the quote and author)

```
def parse(self, response):
    wrappers = response.xpath('//div[@class="post quote"]')
    for wrapper in wrappers:
        quote = wrapper.xpath('span/text()').extract_first()
        author = wrapper.xpath('div[@class="source"]/text()').extract_first()
        yield {'Quote': quote, 'Author': author}
```

```
<div id="entries">
  <div class="post text">...</div>
  <div class="post quote">
    <span>"A book of quotations ... can never be complete."</span>
    <div class="source">—Robert M. Hamilton</div>
    <div class="tags">...</div>
    <div class="post_footer">...</div>
    <!-- Post Footer -->
  </div>
  <div class="post quote">
    <span>"Cheers to a new year and another chance for us to get it right."</span>
    <div class="source">—Oprah Winfrey</div>
    <div class="tags">...</div>
    <div class="post_footer">...</div>
    <!-- Post Footer -->
  </div>
  <div class="post quote">...</div>
  <div class="post quote">...</div>
  <div class="post quote">...</div>
  <div class="post quote">...</div>
</div>
```

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Build the spider

- Steps 1:4, 6:end are the same
- Step 5: parse function

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def parse(self, response):  
    wrappers = response.xpath('//div[@class="post quote"]')  
    for wrapper in wrappers:  
        quote = wrapper.xpath('span/text()').extract_first()  
        author = wrapper.xpath('div[@class="source"]/text()').extract_first()  
        yield {'Quote': quote, 'Author': author}
```

Mini Group Project (MGP) AS2

- Visit <https://quotes.toscrape.com/>
- Create the third spider in this `quote` project to scrape all the quotes, authors, and each quote's first tag on the first page, using wrapper structure
- Save it in a CSV file

